independently activated by its own M button. When either M button is pressed, the joint memory (with the aid of the random number generator) sends a signal to the corresponding apparatus and produces an outcome.

Can this device simulate the quantum mechanics of the two-spin system? Yes, it can—as long as the cable connecting the computers is not disconnected, and as long as it can send messages instantaneously. But unless the system is in a product state and stays in a product state, disconnecting the two computers will destroy the simulation.

Can we prove this? Again, the answer is yes—and that is the essential content of Bell's theorem. Any classical simulation of quantum mechanics that tries to spatially separate Alice's and Bob's apparatuses must have an instantaneous cable connecting the separate computers with a central memory that stores and updates the state-vector.

But doesn't this mean that locality-violating information can be sent through the cable? It would, if Alice, Bob, and Charlie were allowed to do anything that nonrelativistic classical systems can do.[9] But if the only operations that are allowed are those that simulate quantum operations, then the answer is no. As we've seen, quantum mechanics does not allow Alice's density matrix to be affected by Bob's actions.

This problem is not a problem for quantum mechanics. It's a problem for *simulating* quantum mechanics with a classical Boolean computer. That's the content of Bell's theorem: The classical computers have to be connected with an instantaneous cable to simulate entanglement.

7.11 ENTANGLEMENT SUMMARY

Of all the counterintuitive ideas quantum mechanics forces upon us, entanglement may be the hardest one to accept. There is no classical analog for a system whose full state description contains no information about its individual subcomponents. Nonlocality is surprisingly difficult to even define. The best way to come to terms with these issues is to internalize the mathematics. What follows is a compact summary of what we've learned about entanglement. In particular, we've tried to map out the differences